# Change Orders

Federal contract cost growth



**Overview**

**Contracts**

$10K+, Government of Canada

**Awarded**

Apr 2017 – Mar 2023

**Follow-up**

3 years of amendments

**Scope**

All contracts

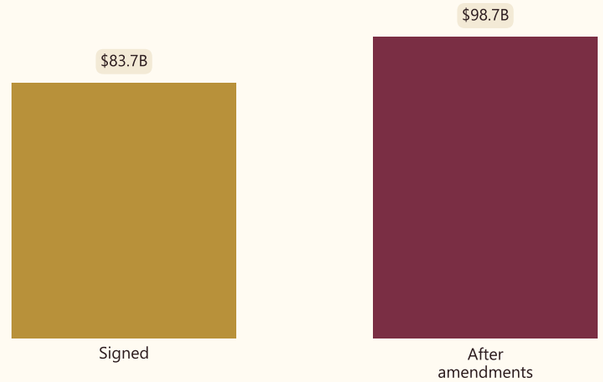
Analysis and design by

**Osman Zakaria, PMP**

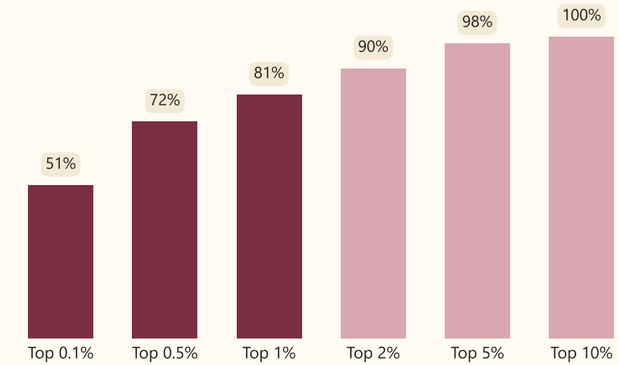
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| 351,872 | 13% | 7.7% | 6.2% | $15.0B | +18% |
|---|---|---|---|---|---|
| Contracts followed | Amended within 3 years | Grew 25%+ | End date pushed 90+ days | Added after signing | Growth on signed value |

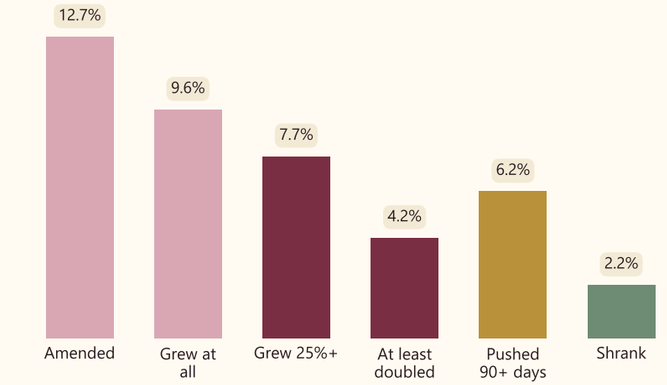
## Value at signing vs after amendments ($B)



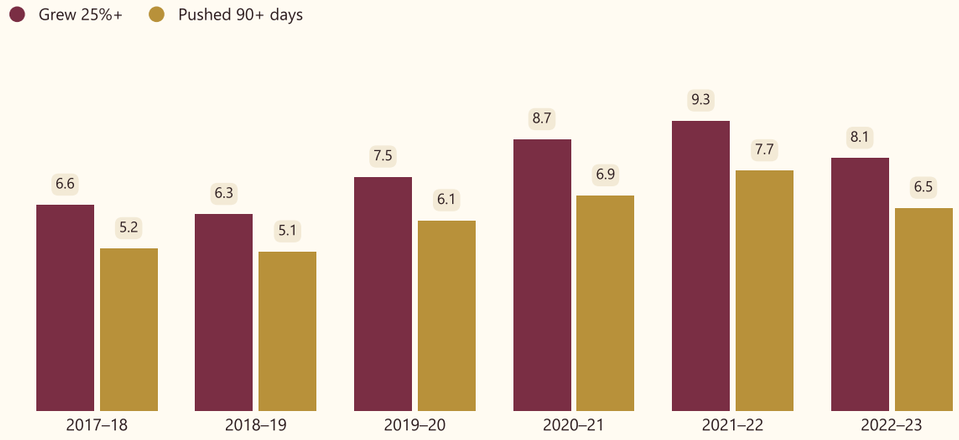

## Growth is concentrated: share of dollars added by the top contracts




## What happens after signing (% of contracts)




## By award year: share growing 25%+ and pushed 90+ days (%)




## Dollars added after signing, by award year ($B)




Source: open.canada.ca, Proactive Publication – Contracts over $10,000; awards April 2017 – March 2023 followed for 3 years. Independent analysis of public data. Not produced by or affiliated with the Government of Canada.

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## What drives growth

**Contracts**
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All contracts

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**12.0%** Services growing 25%+

**2.1%** Goods growing 25%+

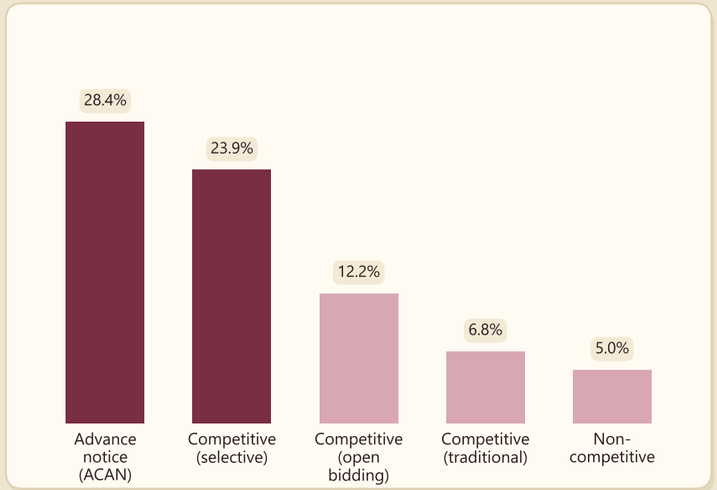
**28%** ACAN sole-source growing 25%+

**19%** 1–3 year contracts growing 25%+

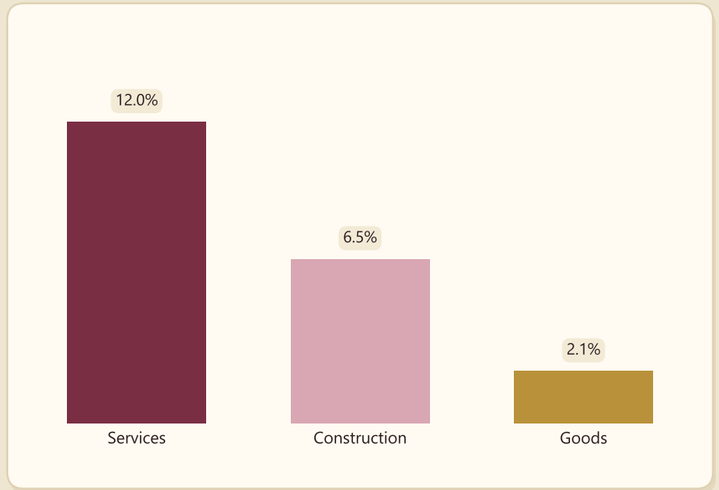
**25%** $10M+ awards growing 25%+

**8.6%** March awards growing 25%+ (raw)

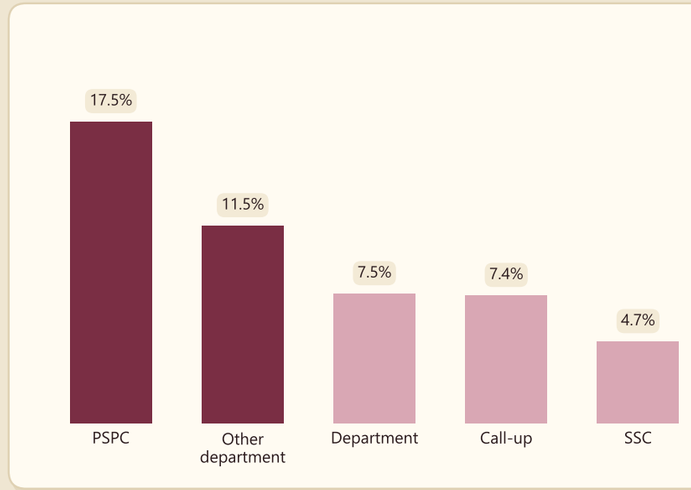
### How it was bought: share growing 25%+ (%)



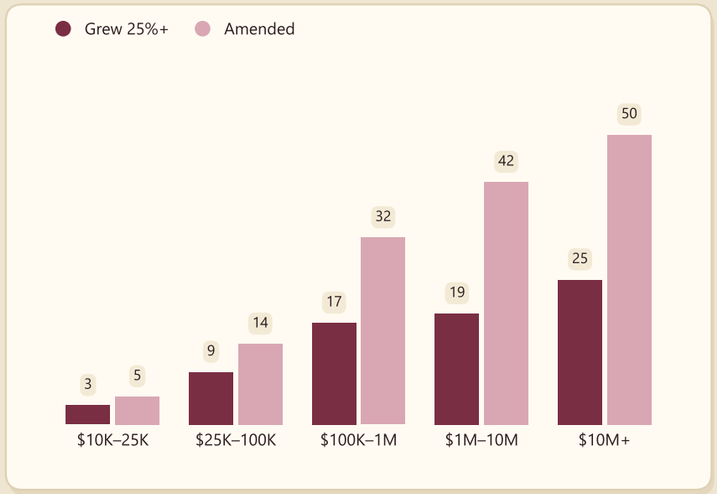

### What was bought: share growing 25%+ (%)



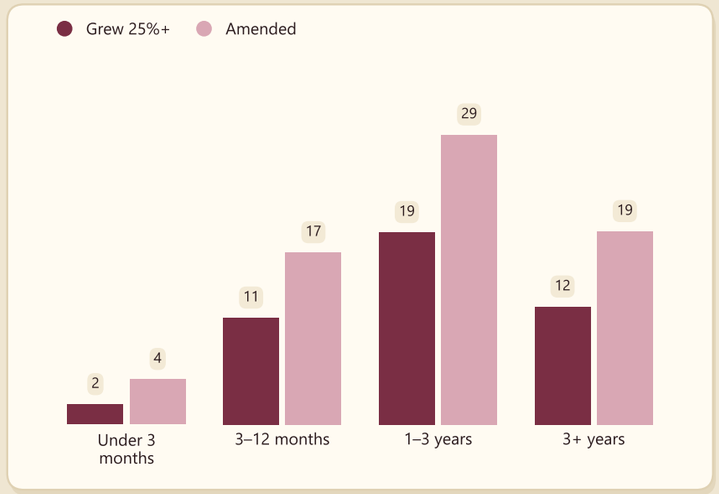

### Who awarded it: share growing 25%+ (%)



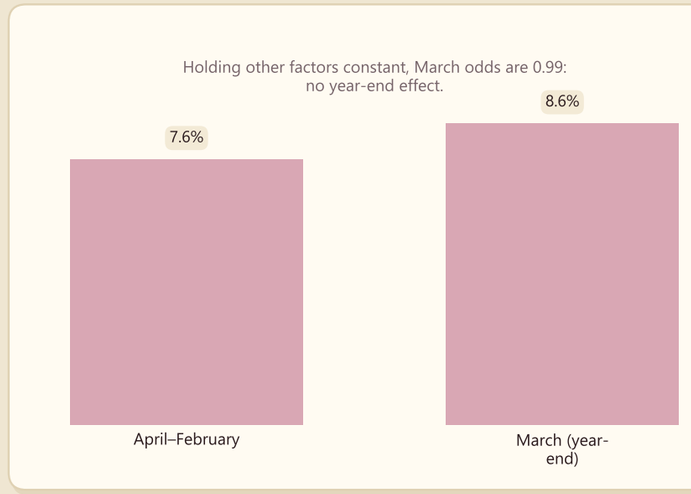

### By award value (% of contracts)




### By planned length (% of contracts)




### Fiscal year-end: share growing 25%+ (%)




Source: open.canada.ca, Proactive Publication – Contracts over $10,000; awards April 2017 – March 2023 followed for 3 years. Rates are raw shares of contracts growing 25%+; page 3 adjusts for everything else. Independent analysis of public data. Not produced by or affiliated with the Government of Canada.

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## Risk factors

**Contracts**

$10K+, Government of Canada

**Awarded**

Apr 2017 – Mar 2023

**Follow-up**

3 years of amendments

**Scope**

Model, all years

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**4.5x** — Odds, 1–3 yr vs under 3 months

**3.1x** — Odds, $10M+ vs $10K–25K

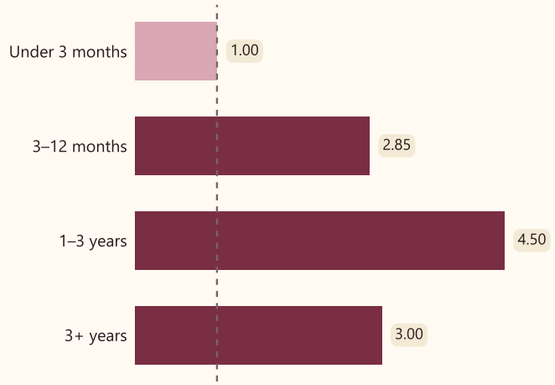
**2.3x** — Odds, awarded by PSPC

**2.1x** — Odds, ACAN sole-source

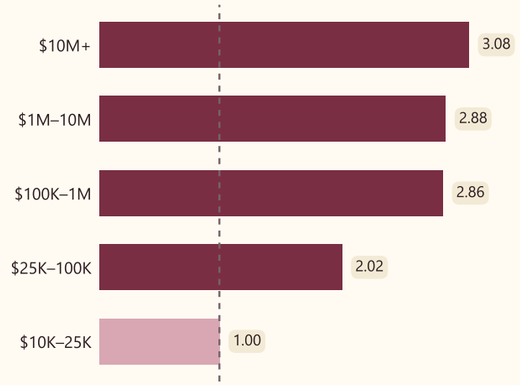
**0.40x** — Odds, goods vs services

**0.99x** — Odds, March awards

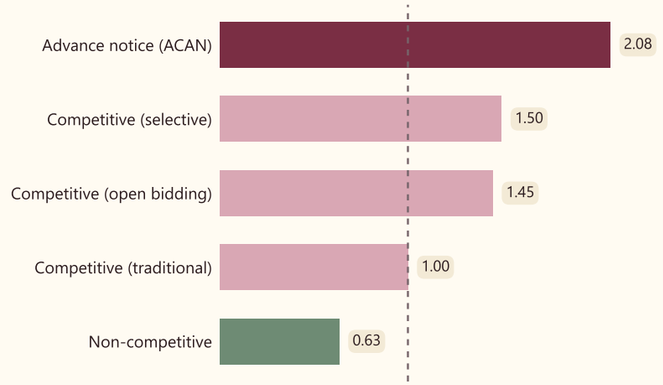
### Planned length (vs under 3 months)



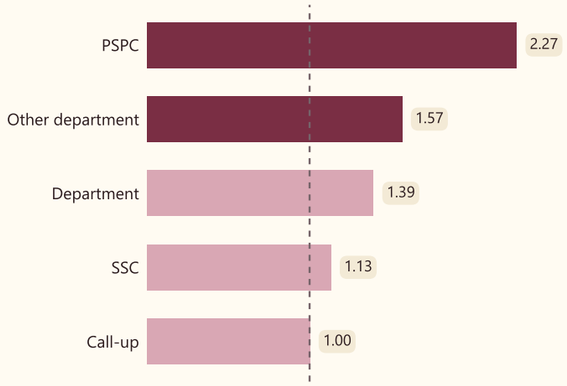

### Award value (vs $10K–25K)




### How it was bought (vs traditional competitive)



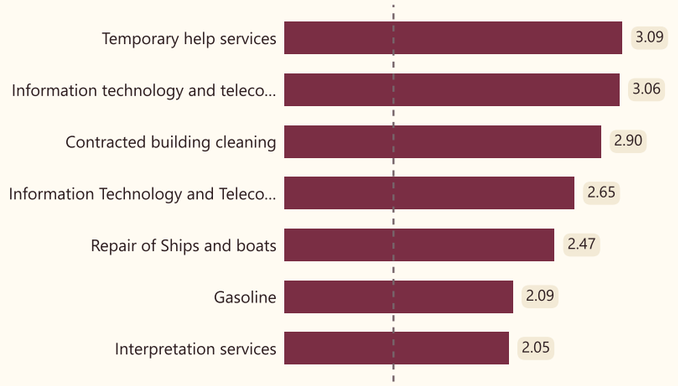

### Who awarded it (vs standing-offer call-up)




### What was bought (vs services)




### Highest-risk spending categories (1,000+ contracts)




Labels as published; the source spells some categories several ways.

Odds ratios of growing 25%+ from an L2-regularised logistic regression on factors known at award, holding the others constant. 1.0 = the reference level (dashed line). Independent analysis of public data. Not produced by or affiliated with the Government of Canada.

# Change Orders

Federal contract cost growth



**Departments**

**Contracts**
$10K+, Government of Canada

**Awarded**
Apr 2017 – Mar 2023

**Follow-up**
3 years of amendments

**Scope**
300+ contracts

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**60** Departments with 300+ contracts

**$2.5B** Added by Shared Services Canada

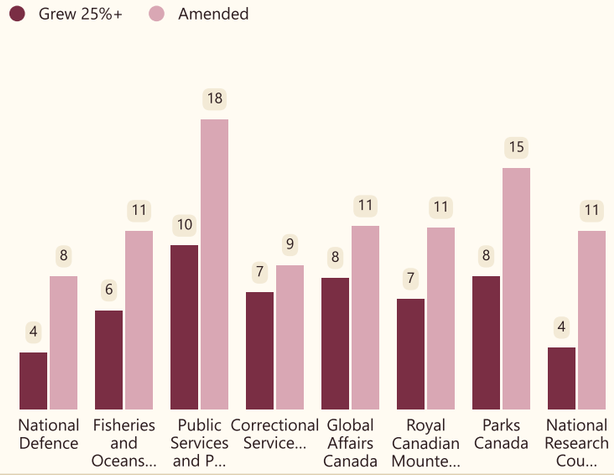
**+24%** Its growth on signed value

**26%** Highest department share growing 25%+

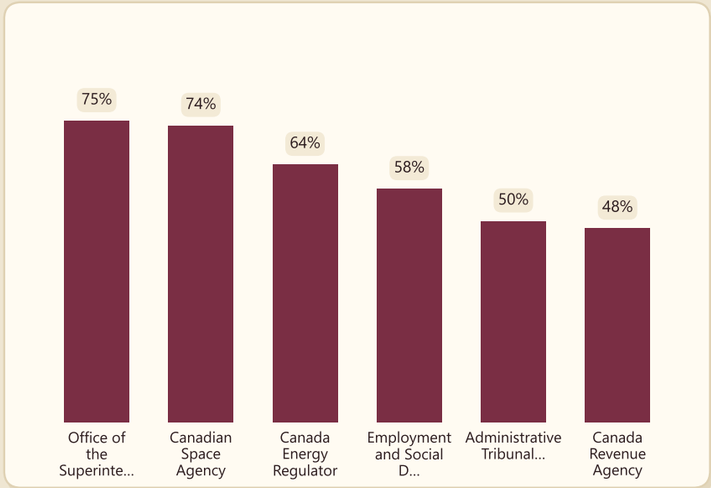
**44%** Of dollars added, top 3 departments

**347,785** Contracts in these departments

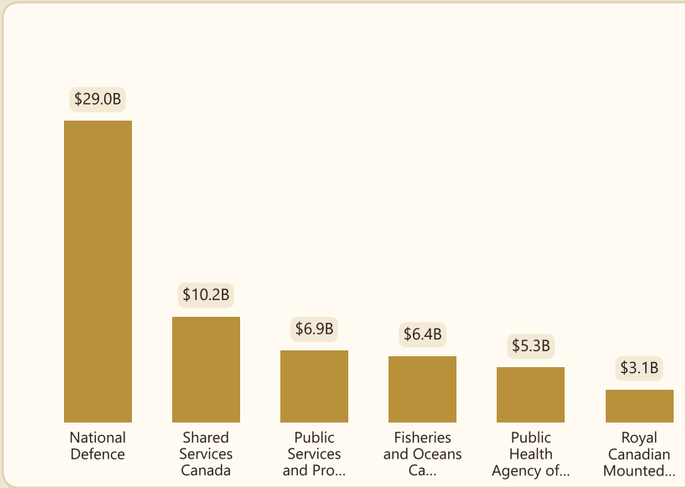
## The 8 busiest departments: growing 25%+ vs amended (%)



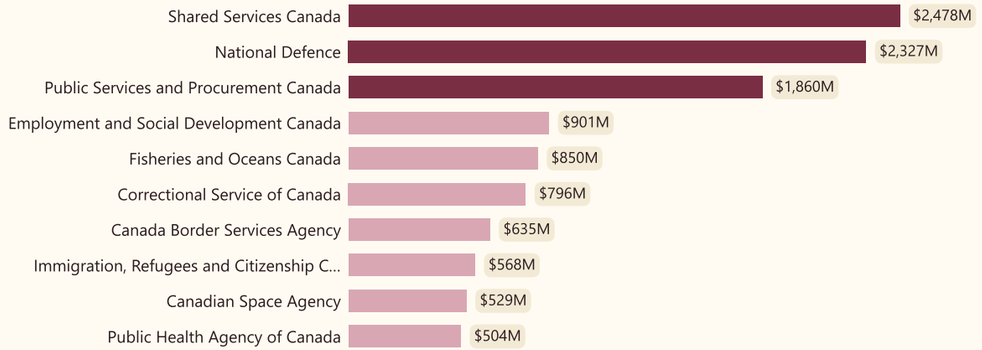

## Highest dollar growth on signed value (%)



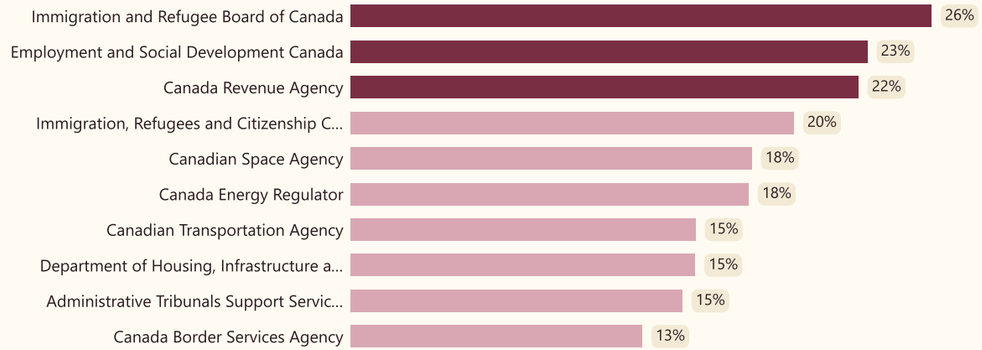

## Largest signed value ($B)




## Dollars added after signing: the 10 largest ($M)




## Highest share of contracts growing 25%+ (%)




Source: open.canada.ca, Proactive Publication – Contracts over $10,000; awards April 2017 – March 2023 followed for 3 years. Departments with fewer than 300 contracts are left out. Independent analysis of public data. Not produced by or affiliated with the Government of Canada.

# Change Orders

Federal contract cost growth



**Risk model**

**Contracts**
$10K+, Government of Canada

**Awarded**
Apr 2017 – Mar 2023

**Follow-up**
3 years of amendments

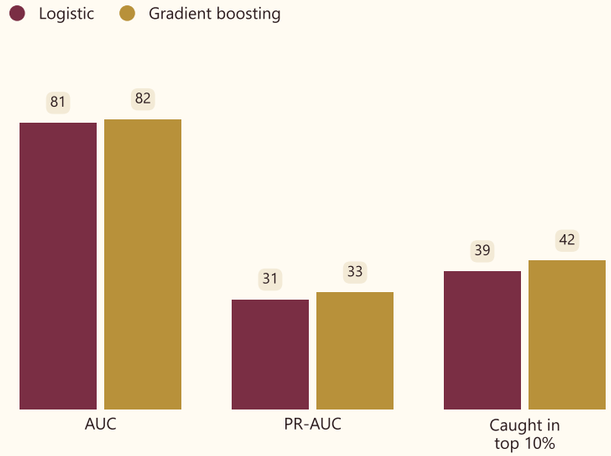
**Scope**
Out-of-time test

1 2 3 4 5 6 7 8 9

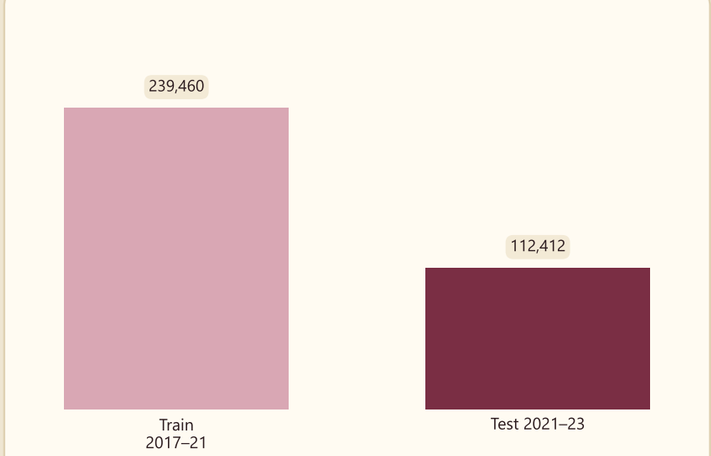
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| 0.81 | 0.82 | 3.9x | 39% | 239,460 | 112,412 |
|---|---|---|---|---|---|
| AUC, out-of-time test | Gradient-boosting benchmark | Lift, riskiest 10% | Of 25%+ growth caught in top 10% | Training contracts | Test contracts, never seen |

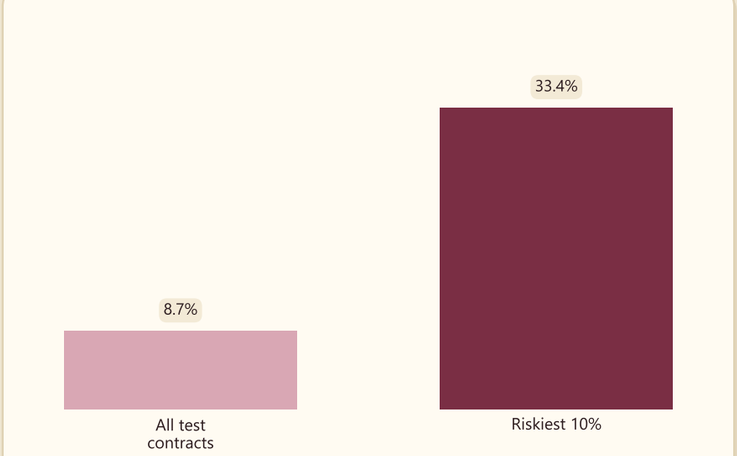
## Logistic model vs gradient-boosting benchmark (%)



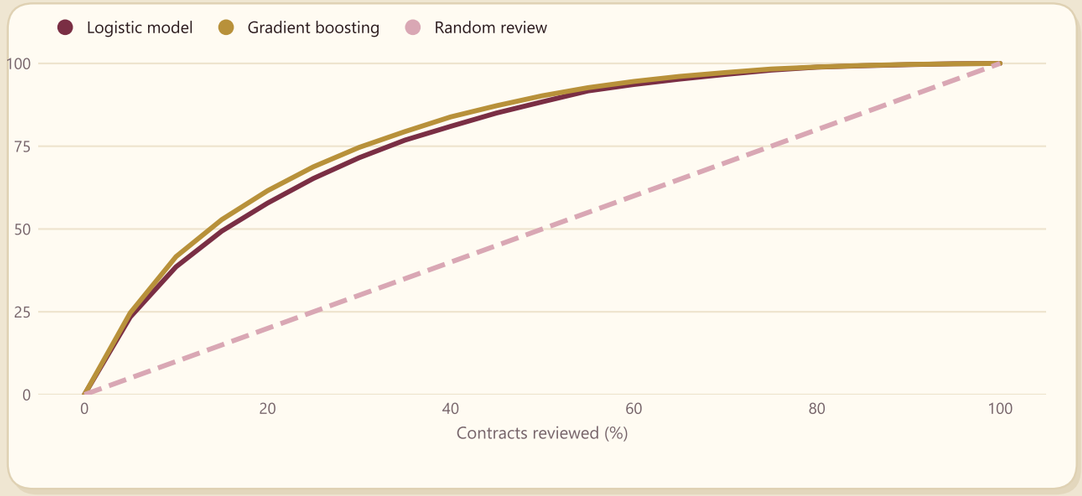

## Contracts in each period



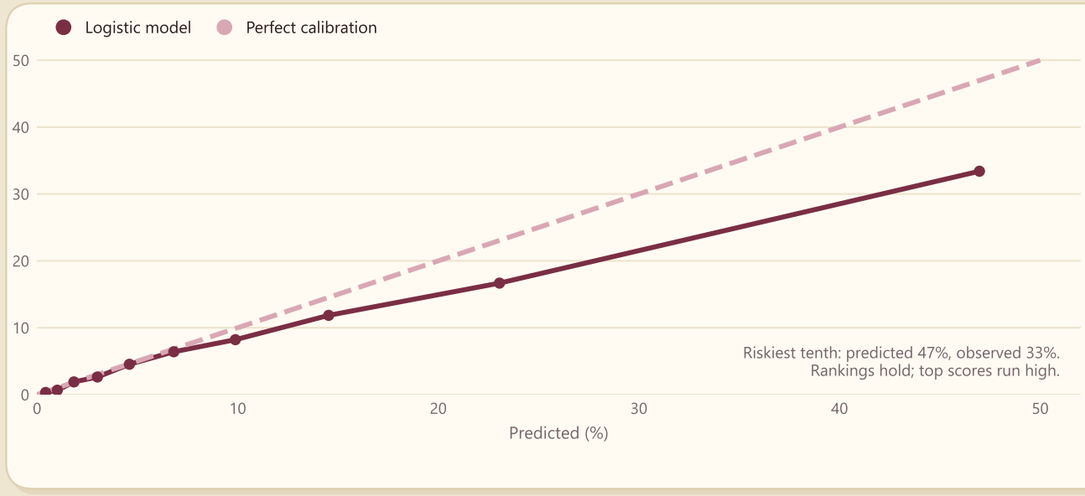

## Base rate vs riskiest tenth: share growing 25%+ (%)




## Gains: share of 25%+ growth caught by reviewing the riskiest contracts (%)




## Calibration: predicted vs observed rate, by tenth of risk (%)




Trained on awards 2017-04 to 2021-03, tested on 2021-04 to 2023-03. The calculator and watch list use the model refitted on all years. Independent analysis of public data. Not produced by or affiliated with the Government of Canada.

# Change Orders

Federal contract cost growth



**Largest & watch list**

**Contracts**

$10K+, Government of Canada

**Awarded**

Apr 2017 – Mar 2023

**Follow-up**

3 years of amendments

**Scope**

$1M+ awards

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**$381M**
Largest single increase

**15.5x**
Its final vs signed value

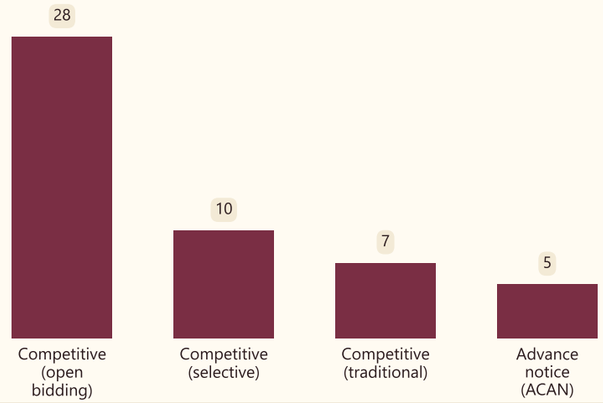
**$4.4B**
Added by the 40 largest

**50**
Awards on the watch list

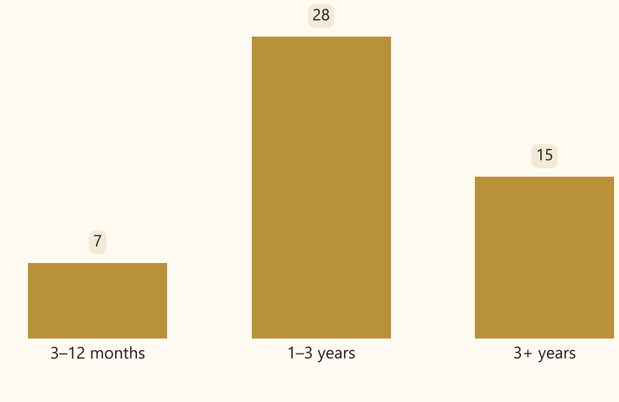
**83%**
Highest predicted risk

**100%**
Of watch list that are services

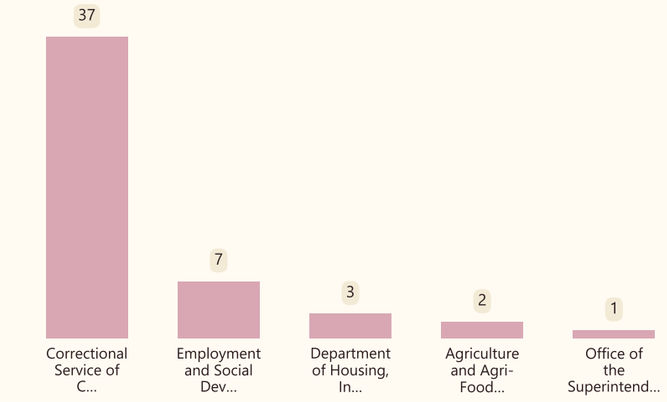
## Watch list by how it was bought (awards)



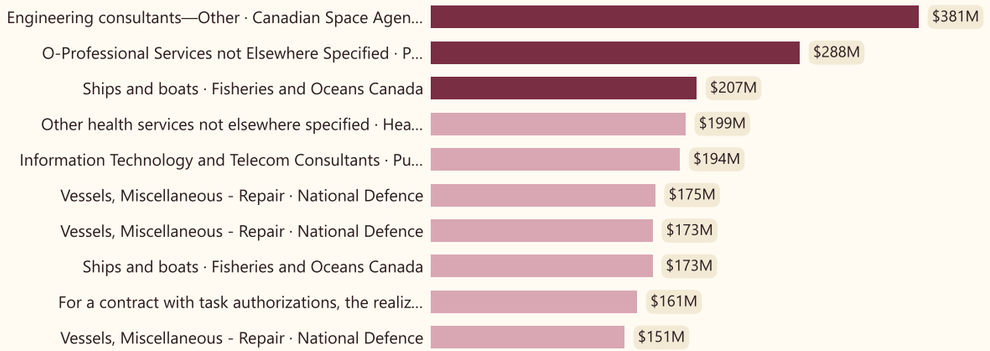

## Watch list by planned length (awards)



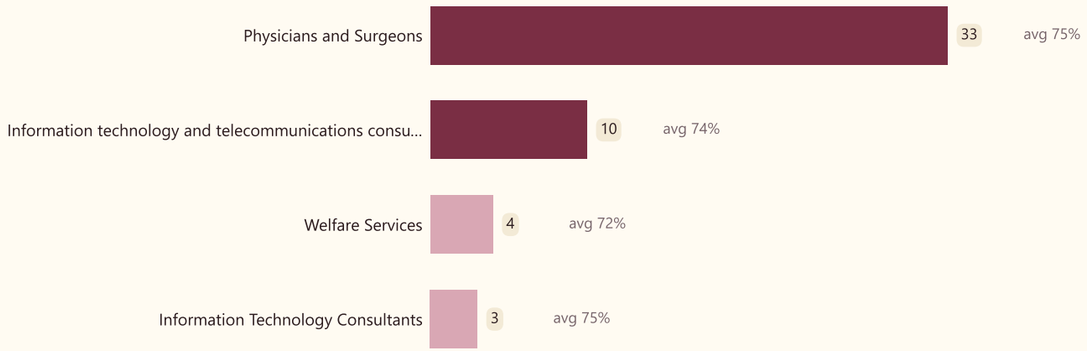

## Watch list by department (top 5, awards)




## The 10 largest increases after signing ($M added)




## Watch list by spending category: awards (average chance of growing 25%+)




Watch list: the 50 highest-risk awards of $1M+ since April 2024, too recent to have an outcome. Suppliers are not named: the score describes the kind of contract, not the supplier. Independent analysis of public data. Not produced by or affiliated with the Government of Canada.

# Change Orders

Federal contract cost growth



## Cost & schedule control

**Contracts**
$10K+, Government of Canada

**Awarded**
Apr 2017 – Mar 2023

**Follow-up**
3 years of amendments

**Scope**
EVM at close

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**$83.7B**
BAC: signed baseline

**$98.7B**
EAC: cost at completion

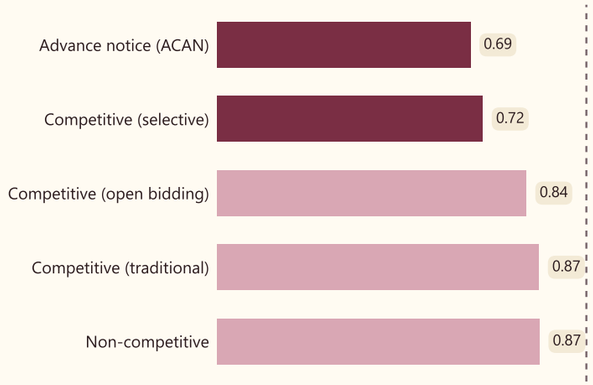
**−$15.0B**
CV = BAC − EAC

**0.85**
CPI at close = BAC ÷ EAC

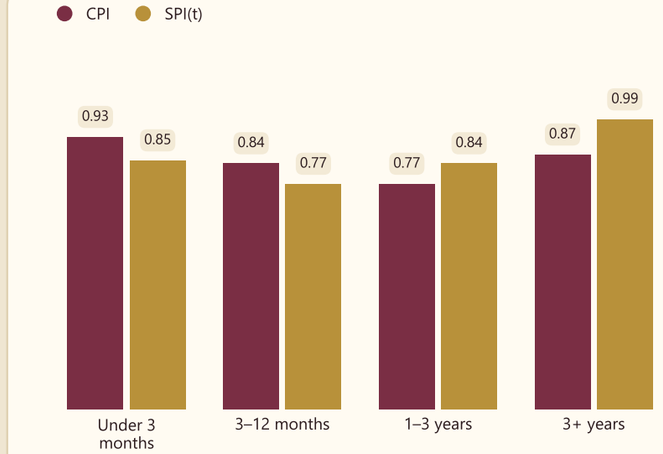
**−18%**
VAC % of baseline

**0.84**
SPI(t) = planned ÷ actual time

### CPI at close by procurement method (1.00 = on baseline)



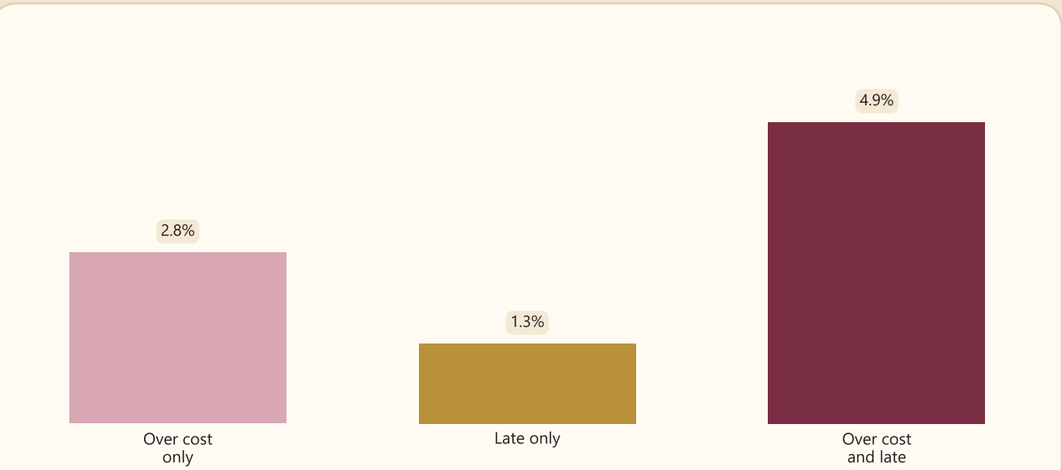

### CPI and SPI(t) at close by planned length



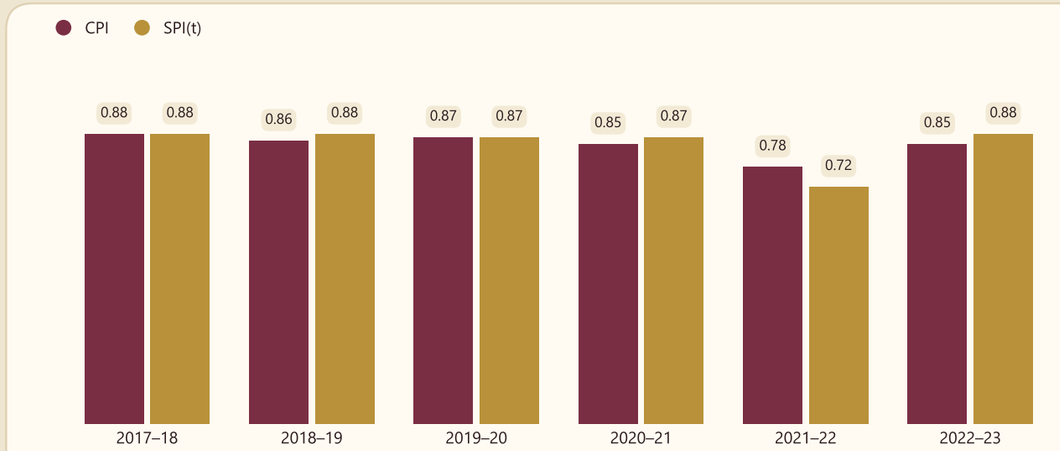

### CPI and SPI(t) at close by commodity




### Off track: 9.0% of contracts (cost 25%+ or 90+ days late), by type (%)




### Portfolio CPI and SPI(t) at close by award year




Earned-value measures at close: each contract is a project, signed value = BAC, value after 3 years of amendments = EAC. No progress data is published, so in-flight CPI and SPI cannot be measured; 1.00 = on baseline. Independent analysis of public data. Not produced by or affiliated with the Government of Canada.

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Federal contract cost growth



**Estimating**

**Contracts**

$10K+, Government of Canada

**Awarded**

Apr 2017 – Mar 2023

**Follow-up**

3 years of amendments

**Scope**

$1M+ reference classes

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**10**
Reference classes, $1M+ contracts

**+69%**
P80 uplift, services 1–3 yrs

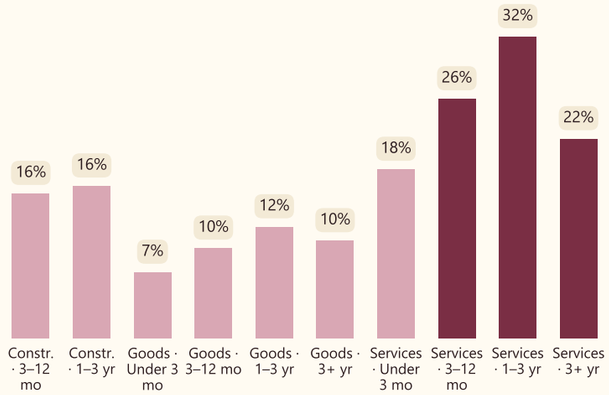
**+0%**
Lowest P80 uplift (goods)

**1.22**
PERT mean, services 1–3 yrs

**±0.22**
PERT σ, services 1–3 yrs

**$12M**
PERT estimate, $10M award

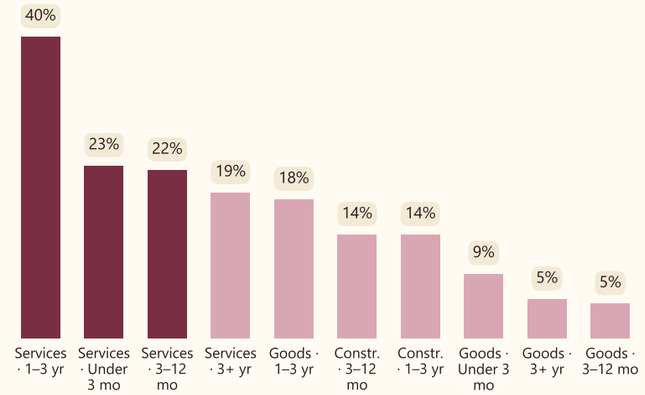
## Chance a major contract grows 25%+, by class (%)



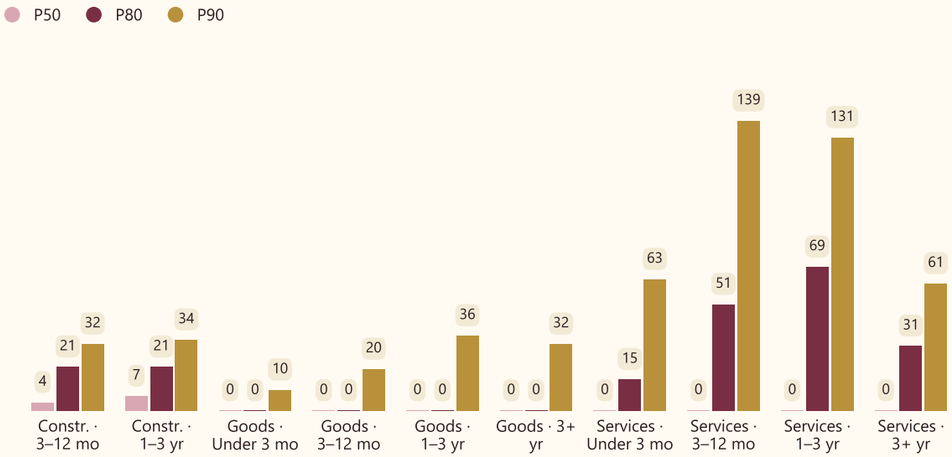

## Worked example: $10M services contract, planned 1–3 years

| | | |
|---|---|---|
| Optimistic (P10) | 1.00 × BAC | **$10M** |
| Most likely (median) | 1.00 × BAC | **$10M** |
| Pessimistic (P90) | 2.31 × BAC | **$23M** |
| PERT mean (O+4M+P)/6 | 1.218 × BAC | **$12M** |
| PERT σ (P−O)/6 | 0.218 × BAC | **$2M** |
| Budget at P80, PERT normal | mean + 0.84σ | **$14M** |
| Budget at P80, reference class | BAC × 1.69 | **$17M** |

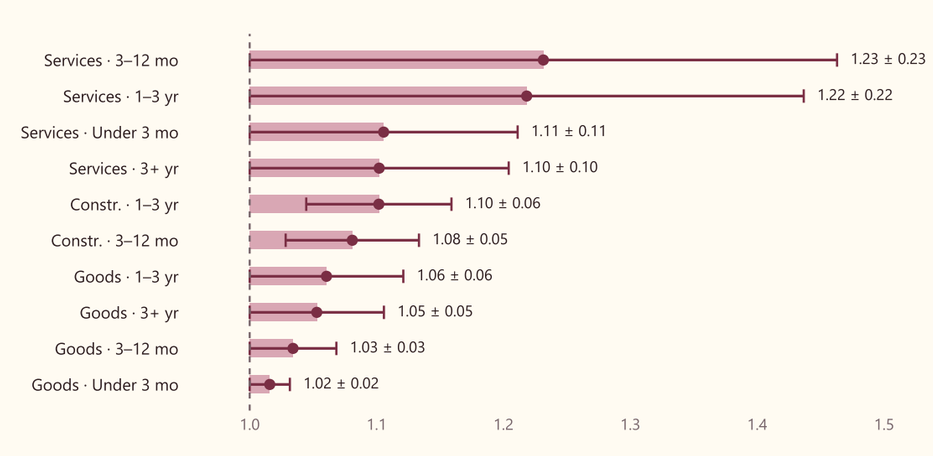
## Dollar-weighted growth on baseline by class (%)




## Reference-class uplift on the baseline for P50 / P80 / P90 confidence (%)




## PERT three-point estimate by class: final ÷ signed value (mean ± σ)




Reference-class forecasting (Flyvbjerg): uplift on the signed baseline needed for 50/80/90% confidence of no overrun, from 2017–23 contracts of $1M+. PERT: O, M, P = P10, median, P90 of final ÷ signed value; mean = (O+4M+P)/6, σ = (P−O)/6. Independent analysis of public data. Not produced by or affiliated with the Government of Ca

# Change Orders

Federal contract cost growth



## Risk & contingency

**Contracts**

$10K+, Government of Canada

**Awarded**

Apr 2017 – Mar 2023

**Follow-up**

3 years of amendments

**Scope**

$1M+ since Apr 2024

1 2 3 4 5 6 7 8 9

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**3,385** — Recent $1M+ awards simulated

**$53.4B** — Their combined BAC

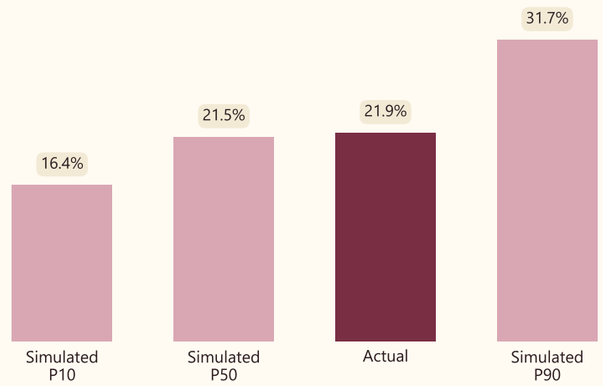
**20%** — Expected cost growth

**26%** — P80 contingency reserve

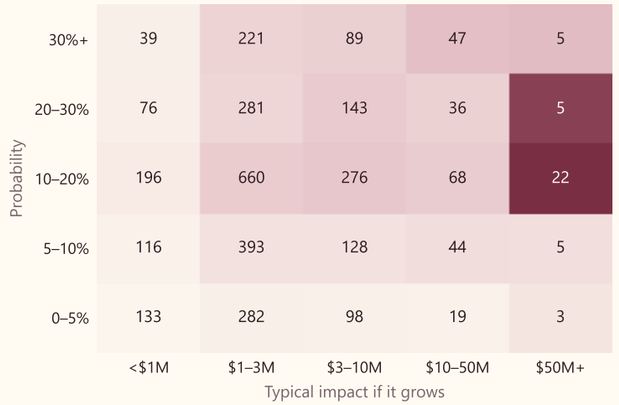
**53th** — Backtest: actual's percentile

**$6.3B** — Total EMV of cost-growth risk

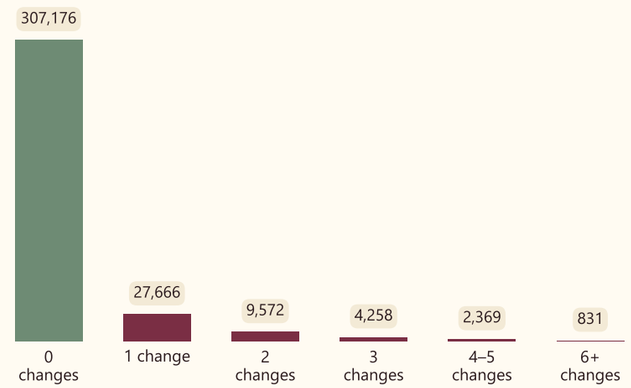
### Backtest: built on 2017–21, run on 2021–23 $1M+ awards (%)



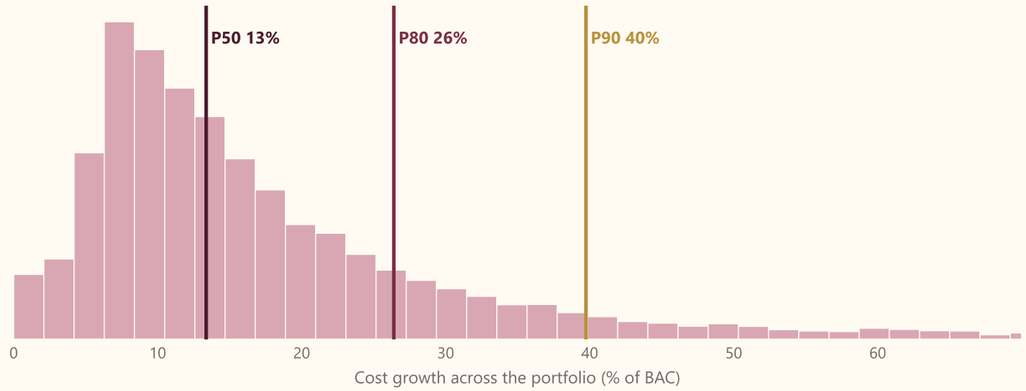

### Probability-impact matrix: awards (shade = EMV)



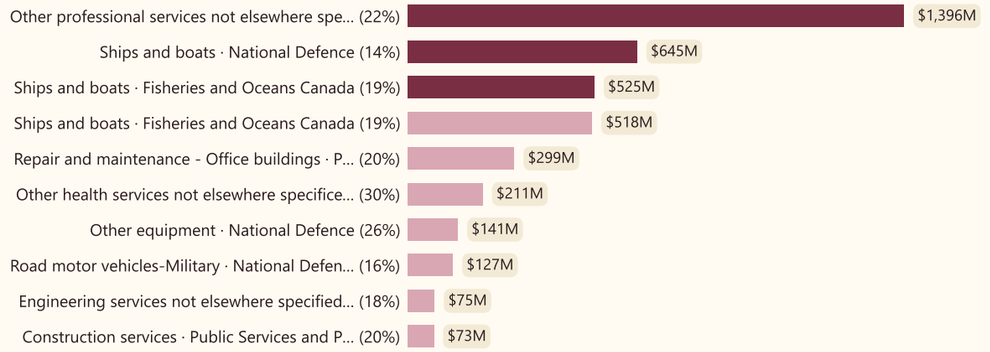

### Change control: amendments per contract (median first change 244 days)




### Simulated portfolio cost growth, % of BAC (10,000 runs)




### Risk register: the 10 highest-EMV awards ($M; probability in brackets)




Monte Carlo (10,000 runs), $1M+ awards since Apr 2024: each grows 25%+ with its calibrated probability; change sizes drawn from past $1M+ contracts of the same kind. EMV = probability × typical impact. Independent analysis of public data. Not produced by or affiliated with the Government of Canada.